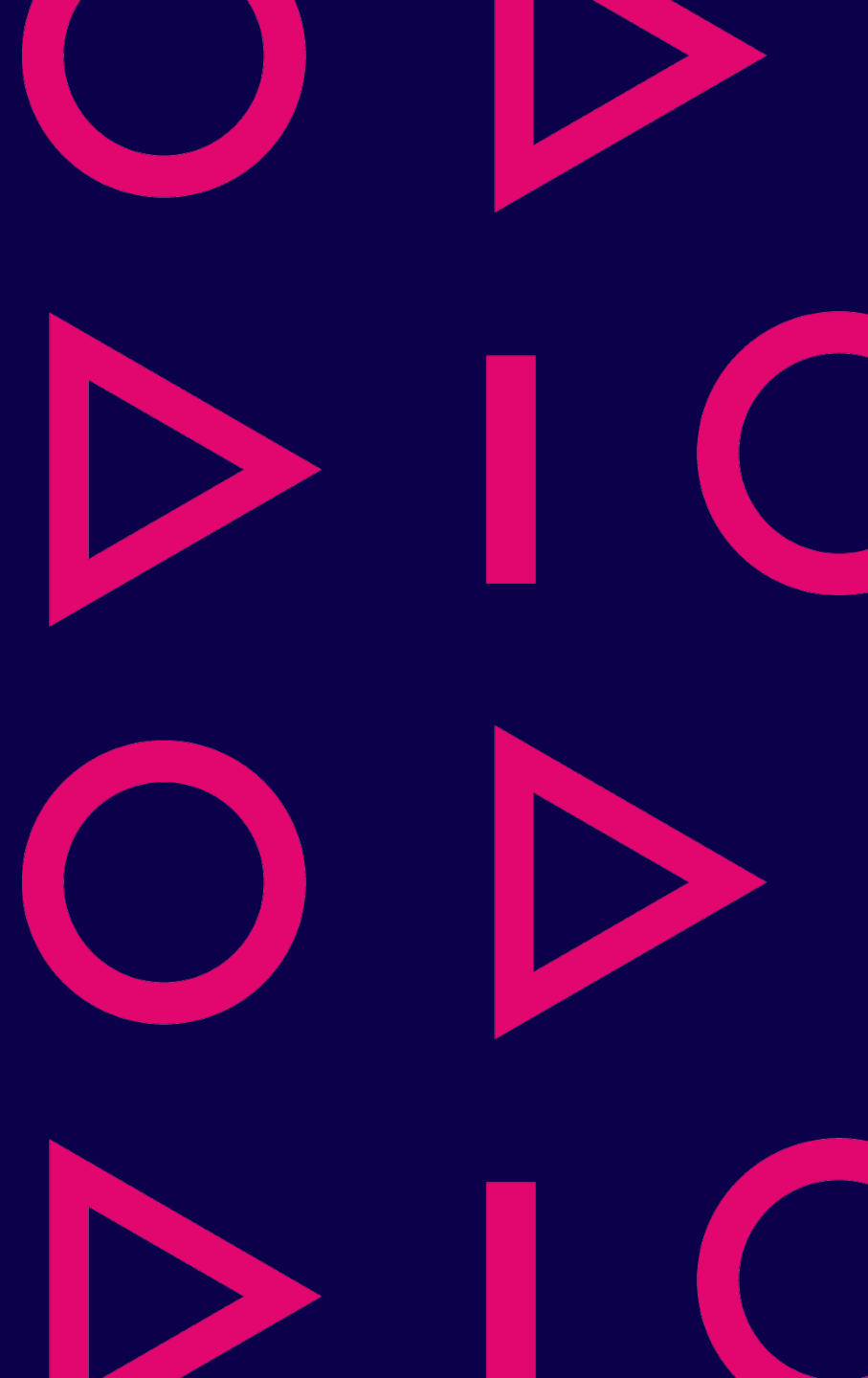


Basics of automation technology

Samppa Alanen

Ref. Teppo Flyktman



CONTENT

- Analog measurements and their implementation in logic programming
- Fieldbus continuous measurements are not processed
- Analog measurement, as well as the processing of analogue information in logic, should be remembered at least at a general level in the exam and when moving on to future courses → Your understanding of this will have time and will have to be increased in future courses, both in practice and in theory.

Analog measurements

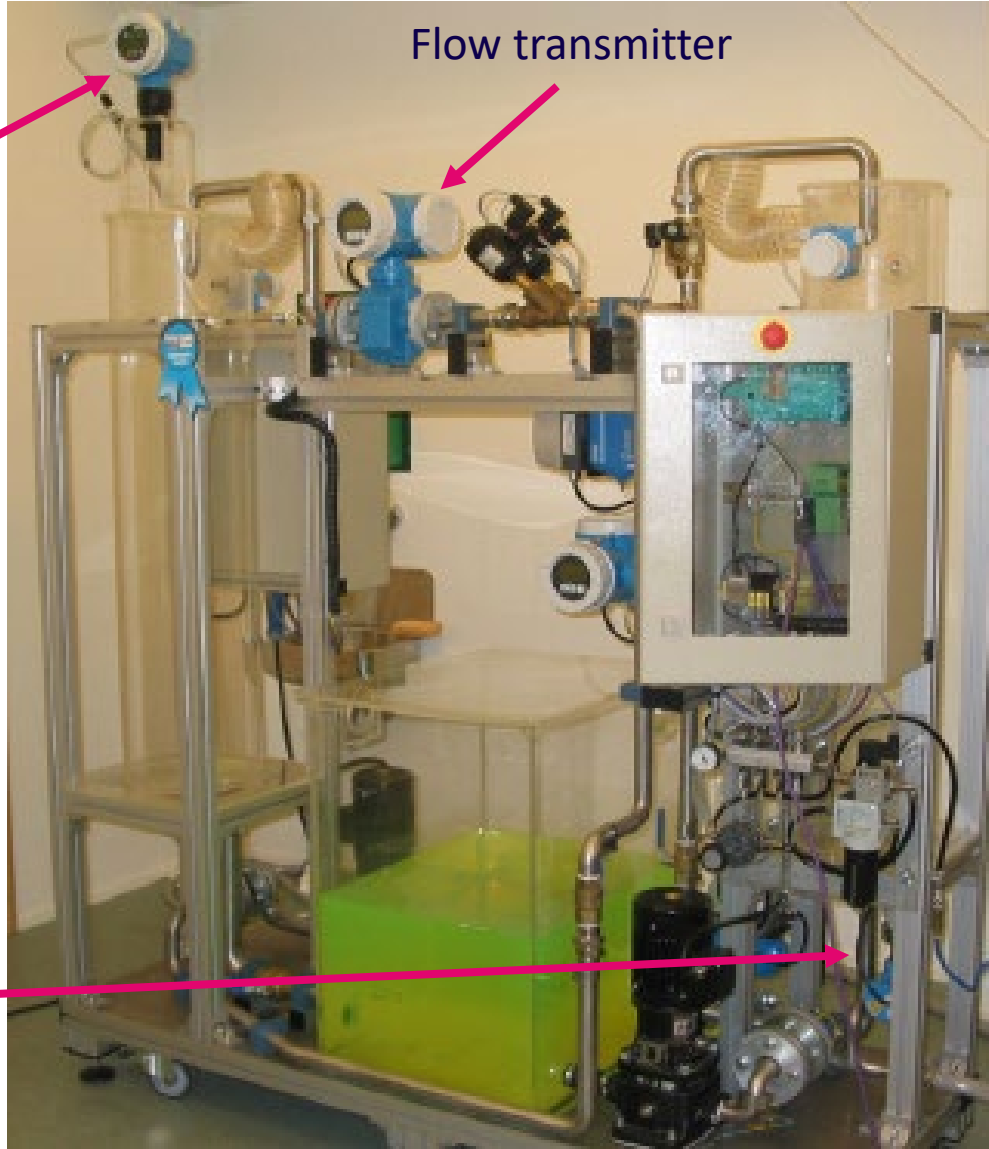
- Continuous measurements
- When measuring the value of something stepless:
 - Temperature
 - Surface level
 - Pressure or differential pressure
 - Flow rate
 - Concentration
- Common in all areas of automation, but especially in process and building automation

Analog measurements

Level measurement
(Ultrasonic)

Level measurement
(pressure)

Flow transmitter

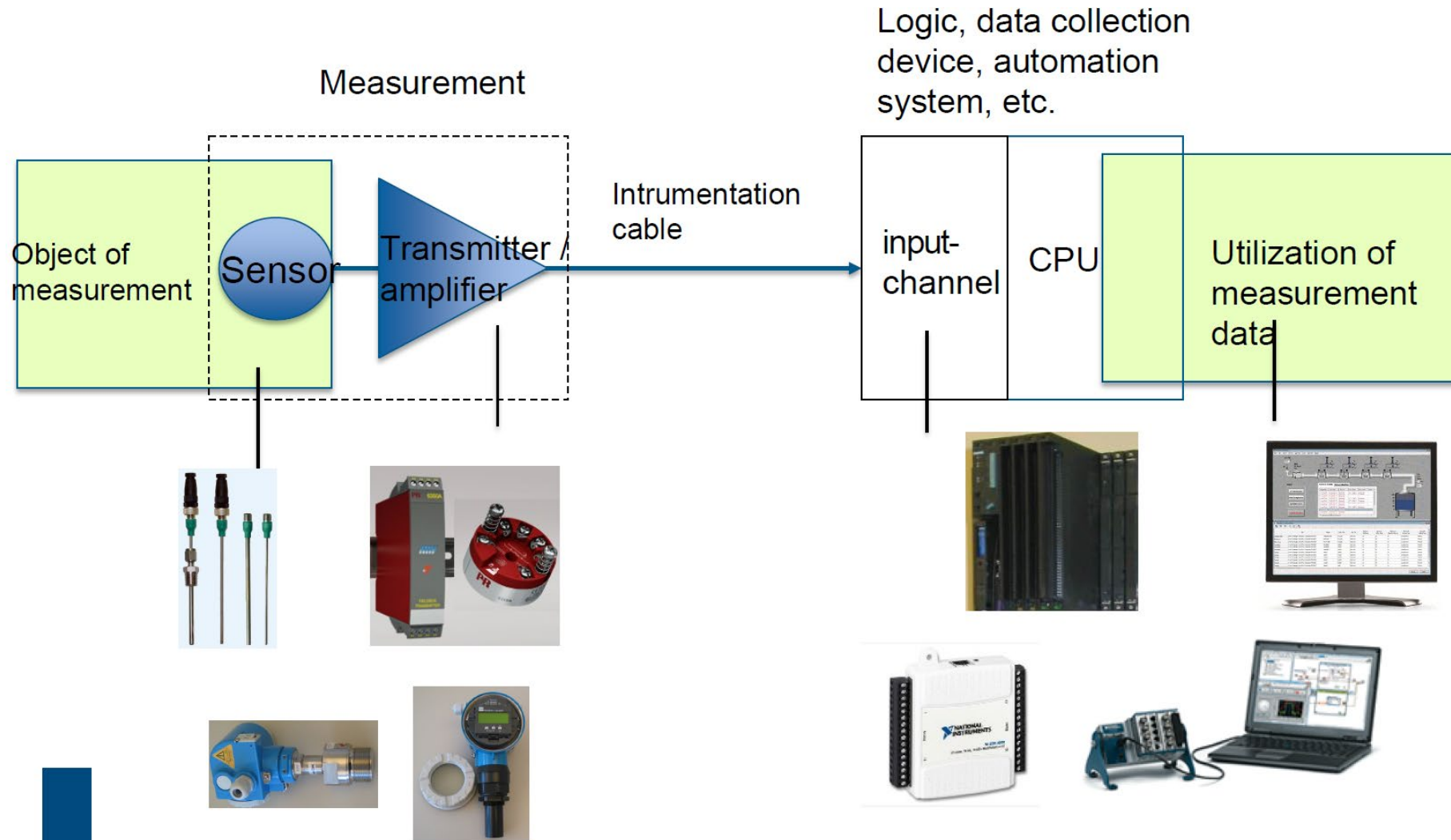


HMI



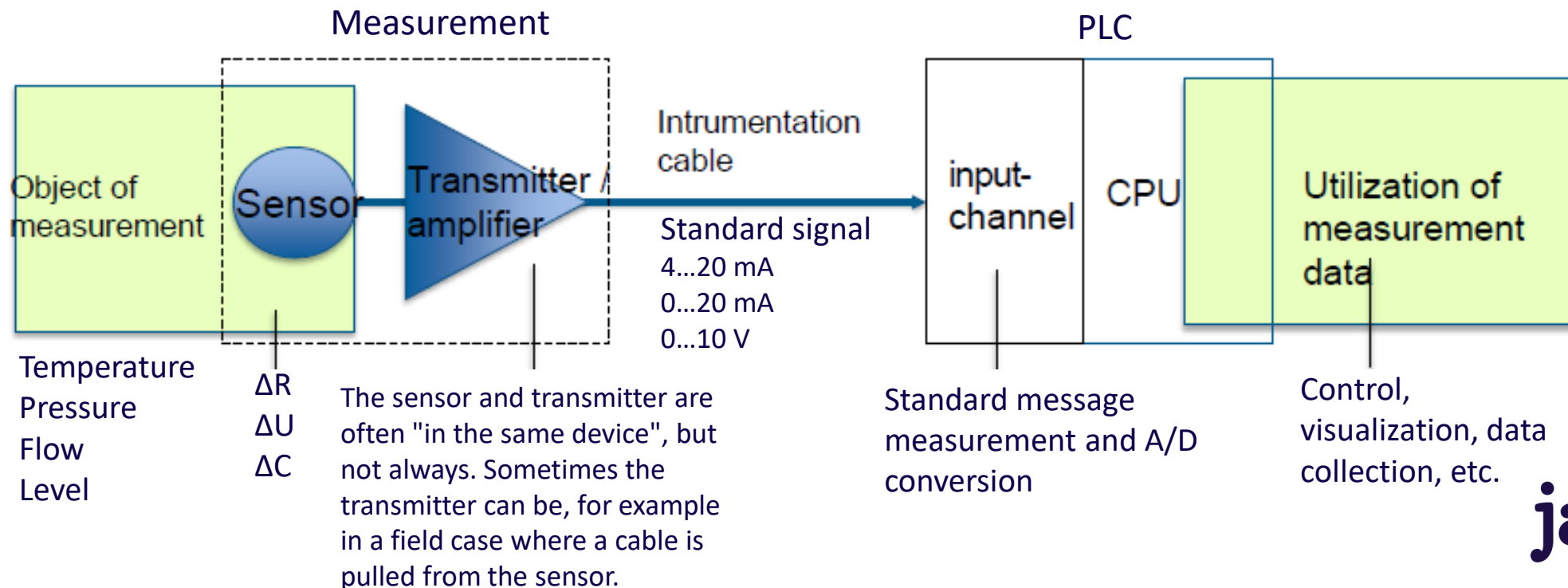
jamk

Analog measurements



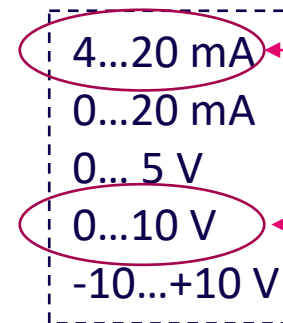
Analog measurements

- **Sensor** Generates a signal proportional to the quantity being measured or modifies the signal fed to the sensor (i.e. generating or modulating)
- **Transmitter** Forms a standard message (mA or V) from the quantity "measured" by the sensor. The term measuring transmitter is often used to refer to the combination of a sensor and a transmitter. E.g. a pressure transmitter with the sensor and transmitter "in the same package".



Standard signals

- To improve compatibility
- Transmits measurement data to a measurement-, data collection- or automation system
- The value to be measured is calculated in the automation or measurement system based on the measurement range corresponding to the message area (must be known!)
- Does not take a position on the quantity to be measured - the same signals for pressure, temperature, level of surface, etc.
- The signals of the measuring instrument and the measuring system must be compatible (same)

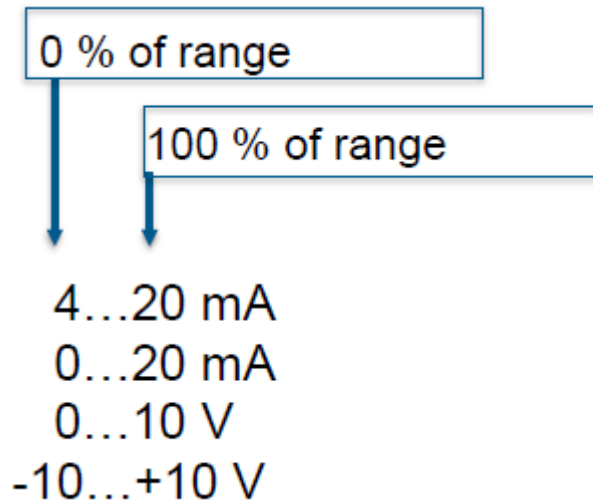


Very common (in industry)

Common (especially in building automation)

Any of these and some others might come across.

Standard signals and measuring range



Example:

The range of a temperature transmitter using a 0...10 V signal is 0°C...100°C

The zero point of the measuring range = 0°C

→ 0V and the maximum point of the measuring range = 100°C → 10V

Measuring range = 100 °C

A 25 °C temperature would correspond to a 2.5 V voltage signal

The measuring device cannot measure values outside the measuring range!

Standard signals and measuring range

Example 2:

The temperature transmitter with 4...20 mA signal has a measuring range of -50°C... 50°C.

Zero point of the measuring range = -50°C

Maximum measuring range = 50°C

Zero point of the measurement signal = 4mA

Measurement signal maximum = 20mA

E.g. a temperature of 25 °C would correspond to a 16mA current message:

$$\frac{25^{\circ}\text{C} + 50^{\circ}\text{C}}{50^{\circ}\text{C} - (-50^{\circ}\text{C})} \times (20\text{mA} - 4\text{mA}) + 4\text{mA} = 16\text{mA}$$

Temperature

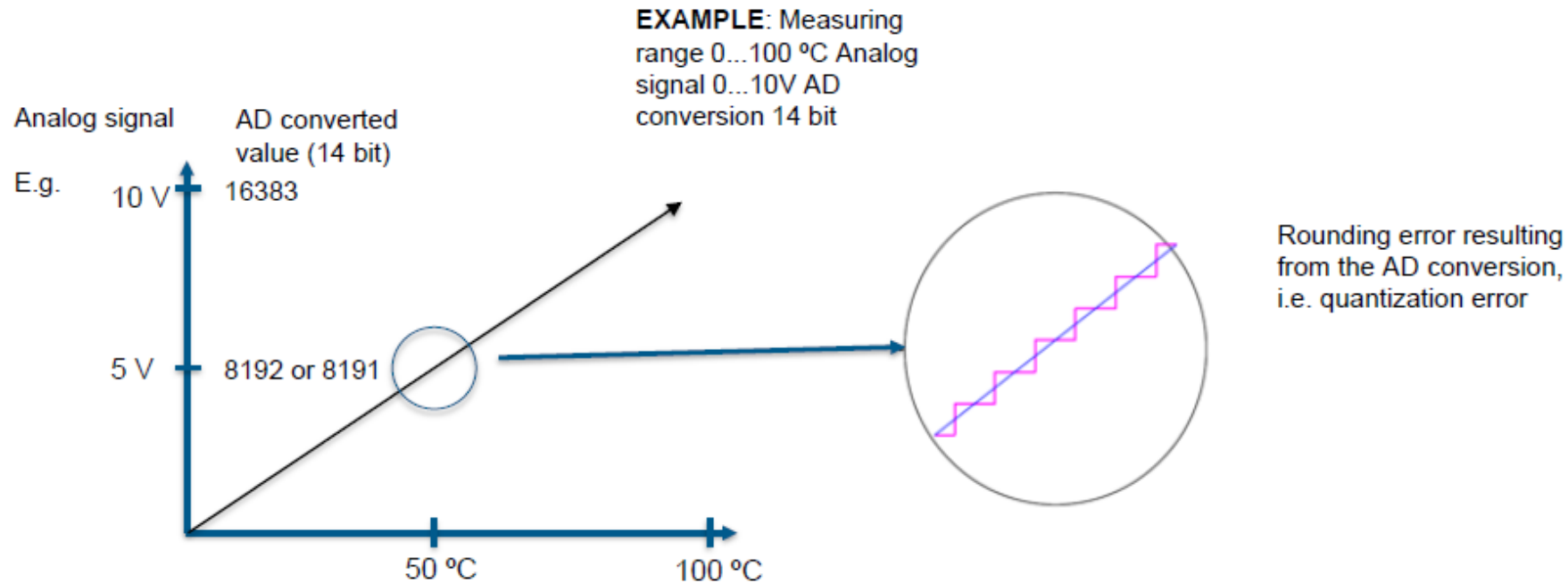
The shift of the zero point is taken into account

Measuring range

Signal range

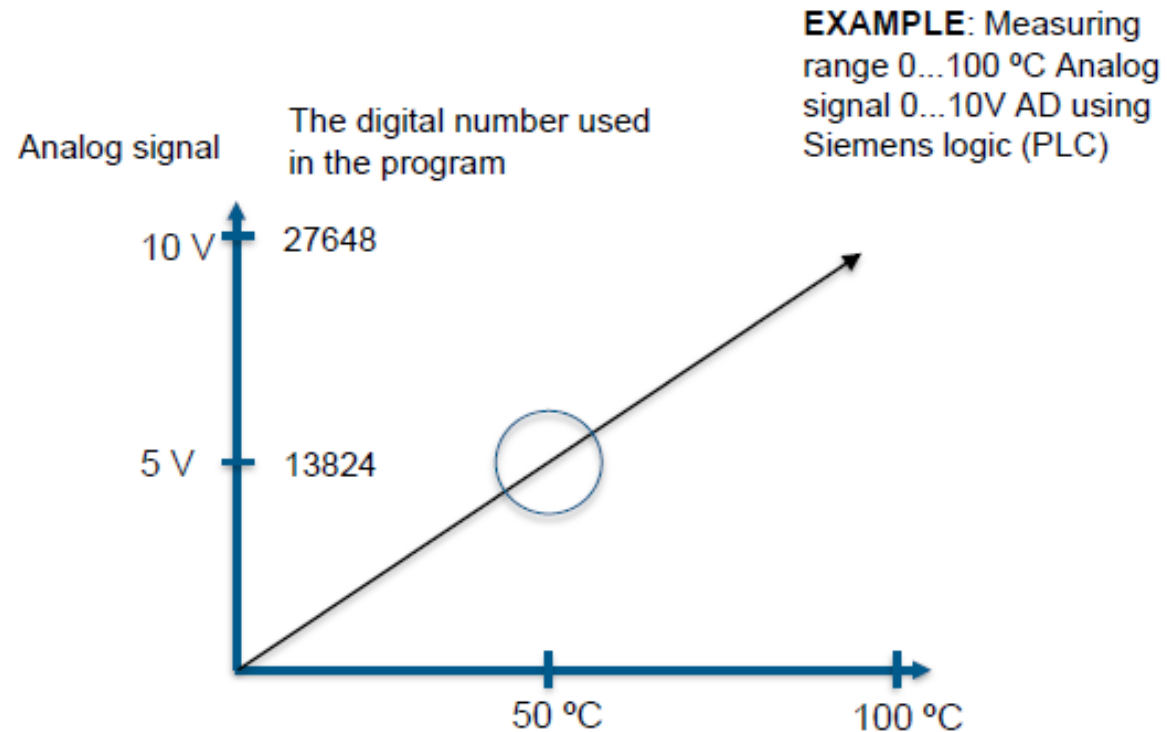
The shift of the zero point is taken into account

A/D conversion and resolution in general



- Digital data can have 2^n different states (n = number of bits)
- Analog information changes "steplessly", but digital information in one-bit-sized "steps"
- Thus, the number of bits in the A/D conversion determines the resolution
- For siemens, see the following slide!

A/D conversion and resolution in general



In the logic program, it is therefore sufficient to know the tuning range of the measuring transmitter and the digital reading range from the AD conversion of the logic in use. The rest is simple arithmetic.

e.g. 0...27648 corresponds to 0...100°C in the example on this page

- When using Siemens logics and their own local analogue IO, the maximum value in the measurement range is equal to 27648. For different logic and IO manufacturers, this figure may be something else! → **You really have to read manual**

Processing an analog value in the program

One way to turn an AD-converted digital number into real-world units:

